Fertility and Housing Tenure Choice

Tommaso De Santo

08/07/2026

Since May

One-market lifecycle model of housing and fertility. What changed:

- **Sequential fertility**: birth-by-birth attempts under declining fertility; children ever born and children at home tracked separately
- **Idiosyncratic income risk**: persistent after-tax earnings process, set externally
- **One housing market**: the two-location structure is set aside for the national question

Model

Environment

- Households live J four-year periods from age 18; retirement, then survival risk
- Two child states: n children ever born, m children currently at home
  - Children at home raise consumption and housing needs
  - Each child matures stochastically, leaves the household, and enters the economy as a young adult
- Idiosyncratic persistent earnings risk; wages and the interest rate are exogenous
- Households rent, or own a discrete house size on a ladder financed with a mortgage
- Government: property tax rebated lump sum; payroll tax funds a pension
- New households enter as young renters: matured children plus an outside inflow

Preferences

Flow utility over consumption and housing services:

$$u(c, \mathbf{h}; m) = \frac{1}{1-\sigma}\left[\frac{c^{\alpha}\left(\mathbf{h} - \underline{h}\, m\right)^{1-\alpha}}{e(m)}\right]^{1-\sigma} + \psi_{\text{child}}\, m.$$

- **Space requirement.** Each child at home absorbs $\underline{h}$ rooms before housing yields services
- **Equivalence scale.** $e(0) = 1$ and $e' > 0$: a larger family needs more resources to reach the same utility
- **Owner premium.** $\mathbf{h} = \chi h$ for owners, with $\chi > 1$; $\mathbf{h} = h^R$ for renters

Bequests

Warm-glow utility at death over terminal wealth:

$$\mathcal{B}(\widetilde{b}) = \theta_0 \frac{(\theta_1 + \widetilde{b})^{1-\sigma}}{1-\sigma}, \qquad \widetilde{b} = b + ph.$$

- θ_0 governs the strength of the bequest motive; θ_1 makes bequests a luxury
- Estates are valued by parents but transfer to no one: no intergenerational wealth flow

Fertility and Child Maturation

- Between ages 18 and 45, households decide each period whether to try for a child: a discrete choice with taste shocks, with separate scales κ_f before and after the first child
- Trying yields a birth with the conception probability

$$\pi(\mathrm{age}) = 1 - \omega_1\, e^{\omega_2(\mathrm{age}-18)}, \qquad \pi = 0 \text{ after } 45,$$

  which falls from 0.98 at 18 to 0.62 at 40: postponed births are increasingly likely never to happen
- Each child lives at home for 18 years on average, with stochastic maturation, raising the equivalence scale and the space requirement
- Matured children enter the economy as young households

Firms and Earnings

- A competitive firm produces with labor, so the wage is pinned down by technology and the labor market clears at any aggregate efficiency units L:

$$Y = A\,L, \qquad w = A$$

- Households supply efficiency units inelastically: an age profile times a persistent idiosyncratic component

$$\log y' = \rho \log y + \varepsilon', \qquad \varepsilon' \sim N(0, \sigma_\varepsilon^2)$$

- Earnings are taxed progressively, a payroll tax funds a stationary pension, and a property tax on owners is rebated lump sum
- With constant returns, a larger population leaves the wage unchanged: housing is the only fixed factor

Housing, Tenure, and Budget Constraints

Households rent a continuous quantity up to a cap, or own a discrete size on a ladder:

$$h^R \in (0, \bar{h}^R], \qquad h \in \{H_1, \ldots, H_K\}.$$

Tenure segmentation: $\max H_k > \bar{h}^R$, so family-sized space is reachable only through ownership.

Renters

$$c + b' + r\, h^R = (1+q)\, b + y(j), \qquad b' \geq 0.$$

Owners, with prior stock h_{-1}

$$c + b' + (\delta + \tau_H)\, ph + \mathbf{1}\{h \neq h_{-1}\}\big[ph - (1-\psi)\, ph_{-1}\big] = (1+q)\, b + y(j), \qquad b' \geq -\phi\, ph.$$

- q: interest rate; δ: maintenance; τ_H: property tax; ψ: transaction cost; $1-\phi$: down-payment share

Housing Supply and Market Clearing

Housing supply slopes upward in levels, and rents map into prices through user cost:

$$H^S = H_0\, r^{\xi}, \qquad r = (q + \delta + \tau_H)\, p.$$

- One market-clearing condition: aggregate housing services demanded equal H^S
- Supply is in levels, not per household: a permanently larger population permanently raises rents
- Stationary equilibrium: the price, the transfer, and the distribution are constant, and the government budget clears

The Household's Problem

Within the period, a household first decides whether to try for a child, then chooses tenure, housing, and saving:

$$V(b,h,y,a,n,m) = \mathbb{E}_\varepsilon \max_{f\in\{0,1\}} \left\{ W^f(b,h,y,a,n,m) + \varepsilon_f \right\},$$

$$W^f = \max_{\text{rent/own},\, h',\, h^{R\prime},\, b'} \; u(c,\mathbf{h};m) + \beta\,\mathbb{E}\big[V(b',h',y',a+1,n',m')\big]$$

subject to the renter or owner budget and the collateral constraint; a successful try ($f=1$, probability $\pi(a)$) adds a child next period.

In old age there are no fertility decisions; survival risk and the warm glow at death close the problem.

Population Closure

Closing the Population

Every version of this model needs a rule for the size of the entering cohort. Three ways to supply one:

1. **Balanced growth path.** Population and all quantities grow at a common rate; the model is read per capita
   - Would need to handle two technical issues here: the committed housing requirement makes preferences non-homothetic, and the housing stock is fixed in levels
   - Counterfactual anyway: the economy is not observed growing along one
2. **Dynamic equilibrium path.** The most general option: no stationarity, solve for the whole time path of the population and prices, markets clearing every period
   - Expensive, and calibration becomes ambiguous: every observed moment must be assigned a date on the path
3. **Stationary equilibrium.** Compare stationary states across policies
   - One long-run number per policy; calibration targets a stationary cross section
   - Needs a rule that pins the level of the population

The stationary comparison is used here.

The Entry Rule

Entrants: matured children who stay, plus an inflow M from outside (immigration). M needs a rule:

$$\text{earlier:} \quad M = M(V) \qquad\qquad \text{now:} \quad M = \bar{M}$$

- **Earlier.** Entry rises with the value V of living in the economy. No moment disciplines that response, and it drives the results: population responses three to ten times larger
- **Now.** A fixed quota: policy moves the population only through births
- **Cost.** Migration cannot respond to a more attractive economy; the earlier numbers bound that channel

Population Balance

The economy has a mass S of households: the population level. In a stationary state entrants replace exits:

$$\underbrace{S\,E_0}_{\text{entrants needed}} = \underbrace{R\,S\,B_0}_{\text{matured children who stay}} + M,$$

with E_0 exits and B_0 maturing children, per household; R the share who stay.

- The native terms scale with S; the inflow M is a level
- As S falls, M covers a growing share of entrants and the decline stops: $S = M/(E_0 - R\,B_0)$
- The outside share of entrants is measured across metro areas in the ACS: 16.9% of young entrant households arrive from outside, in line with national net migration relative to cohort size. Since $s^{out} = 1 - R\,B_0/E_0$, natives replace 83% of each cohort, and M sets the level

Calibration

Externally Calibrated Parameters

	Object	Value	Source
Pref.	σ risk aversion	2.0	standard
Fin.	q interest (annual)	0.04	standard
	δ depreciation (annual)	0.02	Greaney et al.
	τ_H property tax (annual)	0.01	US average
	ϕ financed share	0.80	20% down payment
	ψ transaction cost	0.06	Greaney et al.
Life	17 four-year periods	ages 18–85	retirement 65
Income	ρ, σ_ε, HSV τ	0.914, 0.169, 0.181	FL $\times$ HSV
	τ_{pay} payroll	0.179	Greaney et al.
Children	maturation hazard	2/9 per period	18 expected years
	fecundity ω_1, ω_2	0.02, 0.134	conception schedule
Housing	ladder, rental cap	$\{2, \ldots, 11\}$, 6 rooms	room units

Internally Calibrated Parameters

Param	Description	Estimate
β (annual)	Discount factor	0.9946
ψ_{child}	Child flow utility	0.000
κ_f	Attempt taste scale, first birth	2.987
κ_f^{cont}	Attempt taste scale, later births	1.713
χ	Owner service premium	1.037
H_0	Housing supply intercept	61.706
θ_0	Bequest strength	0.159
θ_1	Bequest shifter	0.094
$\underline{h}$	Child room floor	0.614

Identification

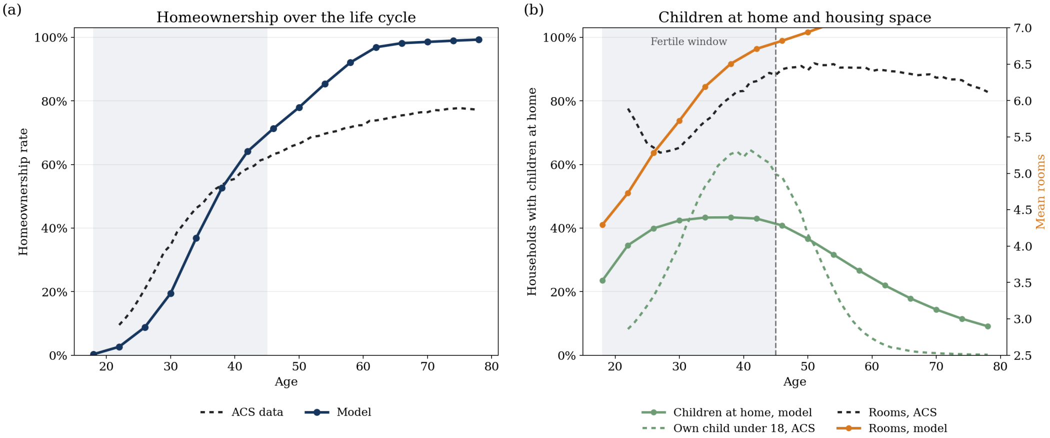
The parameters are chosen jointly, in four economic blocks:

- **Fertility.** The utility value of children ψ_{child} raises the return to every birth: completed fertility. The first-birth noise κ_f spreads the age at which women start trying: mean age at first birth and first births after 30. The continuation noise κ_f^{cont} governs how readily a household that has started keeps going: childlessness
- **Family housing.** The child room floor $\underline{h}$ sets the rooms a child absorbs: the rooms added at the first child, and the rooms difference between families with three or more children and one or two. The family ownership gap has no parameter of its own and enters as an overidentifying restriction
- **Tenure and supply.** The owner premium χ raises the utility of owning over renting: the ownership rate. The supply scale H_0 sets the level of the housing stock: aggregate rooms
- **Saving and bequests.** The discount factor β governs how much wealth households carry through life: net worth over earnings. The bequest strength θ_0, how much is held back to leave at death: the bequest flow. The curvature θ_1, how the motive strengthens with wealth: old-age p90/p50

Model Fit

	Moment	Target	Model
Fertility	Completed fertility	1.918	1.967
	Childlessness	0.188	0.104
	Mean age at first birth	25.31	25.57
	First births at 30+	0.270	0.327
Housing	Rooms response to first birth	0.664	0.690
	Rooms, 3+ vs. 1–2 children	0.368	0.359
	Family ownership gap	0.168	0.162
	Ownership rate	0.575	0.582
	Aggregate mean rooms	5.780	6.478
Wealth	Wealth / earnings	6.873	5.794
	Bequest flow / wealth	0.0088	0.0087
	Old wealth p90/p50	3.448	2.090

Lifecycle Profiles



Funded Property-Tax and Grant Reforms (preliminary)

The experiment: raise the annual property tax from 1% to 2% of house value and return the revenue to households, either as an equal lump-sum rebate or as a grant paid at the purchase of a family-sized home. Percent changes across stationary states:

Policy	TFR	Births	Households	House price	Rent
2% tax, lump-sum rebate	$+0.38$	$+0.33$	$+1.70$	-17.5	-6.1
2% tax, purchase grant	$+0.75$	$+0.67$	$+3.46$	-17.3	-5.8

- The tax lowers the house price, so the down payment falls by 17.5%; rents fall only 6%, because rent includes the tax itself
- In U.S. units, $+0.33$ to $+0.67$% of births is roughly 12 to 24 thousand additional babies per year